

Trading Scope for Credibility in Difference-in-Differences

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Abstract

Difference-in-differences practitioners routinely confront parallel trends failing for some treated cohorts but not others. Existing responses keep the average treatment effect on the treated (ATT) as the target. Heterogeneity-robust estimators point-identify it while maintaining parallel trends, and sensitivity methods bound it, so both are compromised, through bias or a wide set, exactly when some cohorts offend. We propose a different response: change the estimand. We define a credible-subpopulation local ATT (LATT), the treatment effect for the subpopulation of cohorts whose parallel trends assumption is credible, and show it is point-identified precisely when the full ATT is not. Estimation is a reweighting of standard group-time effects toward credible cohorts, combined with honest sensitivity bounds on the residual violation of the selected set. The method’s advantage over the ATT grows with how informative pre-trends are about post-treatment violations. Simulations confirm that the LATT recovers a clean effect where the ATT estimator is biased and that honest sensitivity bounds restore coverage where point estimates collapse. An application to the local house-price effect of the shale boom illustrates the method: a significantly positive pooled estimate is shown to rest entirely on cohorts whose prices were already trending before onset, and the credible subpopulation reveals no house-price effect once these are set aside.

Keywords: Difference-in-differences, Staggered adoption, Parallel trends, Robust inference, Sensitivity analysis, Partial identification.

JEL classification: C1, C23, C51.

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1 Introduction

Difference-in-differences is among the most widely used research designs in empirical economics. Its credibility rests on the parallel trends assumption: absent treatment, the average outcomes of treated and comparison units would have evolved in parallel. Researchers are rarely certain that this holds, and it has become standard practice to assess it by testing for pre-treatment differences in trends.

That practice is fragile. Pre-trends tests are often underpowered against economically meaningful violations (Bilinski and Hatfield, 2020; Freyaldenhoven et al., 2019; Kahn-Lang and Lang, 2020; Roth, 2022). Conditioning the analysis on passing them induces a pre-test selection bias (Roth, 2022). And even a clean pre-period does not guarantee a clean post-period, since nothing prevents a confound from arriving at the moment of treatment (Kahn-Lang and Lang, 2020).

A large recent literature, surveyed in Roth et al. (2023) and de Chaisemartin and D’Haultfoeuille (2023), has responded by relaxing parallel trends itself. Building on Manski and Pepper (2018), Rambachan and Roth (2023) abandon exact parallel trends and instead bound how different the post-treatment violations can be from the observed pre-trends, reporting a uniformly valid confidence set for the ATT in place of a point estimate. A Bayesian variant (Kwon and Roth, 2024) instead places a prior on the violation δ , calibrated from the pre-trends by empirical Bayes, to sharpen the set into a posterior mean and credible interval. Both keep the ATT as the target and use the pre-trends to extrapolate the violation across periods. Our method uses them to select across cohorts. In non-staggered designs with a long pre-period, de Chaisemartin (2026) instead ranks the post-treatment difference-in-differences against the distribution of the pre-treatment ones, replacing parallel trends with a stability restriction.

The ATT is where a structural limitation appears. When parallel trends fails for some cohorts but holds for others, the ATT, an average over all treated cohorts including the offending ones, is precisely the object that is hard to recover. A point estimate of it is then biased, and the sensitivity set, though honest, can be wide and uninformative, because it must accommodate the worst offenders in the average.

This paper takes a different route, familiar from other corners of causal inference: when the population target is not credibly identified, retreat to a subpopulation where it is. This is the logic of the local average treatment effect (Imbens and Angrist, 1994), of the optimal-subpopulation average treatment effect under limited overlap (Crump et al., 2009), and of overlap weighting (Li et al., 2018). In each case one trades the original estimand for a credibly-identified effect on a selected subpopulation, and defends the trade by characterizing whom the subpopulation comprises.

We develop the staggered-DiD instance of this principle. We define the credible-subpopulation LATT, the weighted average of group-time effects restricted to cohorts whose parallel trends assumption is credible, and show that it is point-identified even when the full ATT is not, because the violations of the excluded cohorts do not enter it. Estimation is a transparent reweighting of the group-time effects of Callaway and Sant’Anna (2021), which we adopt for concreteness. Because the method only restricts the aggregation to a subset of cohorts, it applies to any heterogeneity-robust

estimator that delivers group-time effects.¹ The credibility of a cohort is assessed from its pre-trends, so that the reweighting is a data-dependent selection, and we treat inference on the selected set with the sensitivity machinery of Rambachan and Roth (2023).

The method thus occupies a distinct point on a frontier trading off credibility, precision, and scope. Relative to a heterogeneity-robust point estimate of the ATT, it targets a subpopulation to buy back credibility. Relative to a HonestDiD set for the ATT, it recovers a sharper, point-identified answer to a narrower question.

Its advantage is governed by how informative pre-trends are about post-treatment violations. When they are uninformative, selecting on them buys nothing and the identifying assumption must rest on economic context. In that regime the natural relative-magnitudes sensitivity restriction is self-undermining. Anchoring its bounds to the observed pre-trends, it reports false precision when those are flat, whereas a fixed restriction, whose bound is set by the researcher rather than read off the pre-trends, is not.

The remainder of the paper is organized as follows. Section 2 fixes the staggered-DiD setup, defines the credible-subpopulation LATT, and develops its identification, estimation, and sensitivity-based inference. Section 3 reports the simulation study, Section 4 applies the method to the local house-price effect of the shale boom, and Section 5 concludes.

2 The credible-subpopulation LATT

2.1 Setup

We adopt the staggered adoption framework of Callaway and Sant’Anna (2021). There are periods $t = 1, \dots, T$ and units i indexed by their first treatment date $G_i \in \{2, \dots, T\} \cup \{\infty\}$, where $G_i = \infty$ denotes never-treated. Let $Y_{it}(g)$ denote the potential outcome in period t if unit i is first treated in period g , and $Y_{it}(\infty)$ the never-treated potential outcome. The building-block causal parameter is the group-time average treatment effect on the treated,

$$\text{ATT}(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g]. \quad (1)$$

Summary parameters are weighted aggregations $\theta = \sum_g \sum_t w(g, t) \text{ATT}(g, t)$ with researcher-chosen weights (Callaway and Sant’Anna, 2021), of which the overall ATT and the event-study path are special cases.

It is convenient to work with the event-study representation and the decomposition of Rambachan and Roth (2023). Let $\hat{\beta} = (\hat{\beta}'_{\text{pre}}, \hat{\beta}'_{\text{post}})'$ be an asymptotically normal vector of event-study coefficients estimated by any of the heterogeneity-robust procedures above, with $\sqrt{n}(\hat{\beta} - \beta) \rightarrow \mathcal{N}(0, \Sigma)$. The

¹These estimators (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024; de Chaisemartin and D’Haultfœuille, 2020) were developed to repair the aggregation failure of two-way fixed effects under staggered adoption, whose implicit weights on group-time effects can turn negative when treatment effects are heterogeneous (Goodman-Bacon, 2021; de Chaisemartin and D’Haultfœuille, 2020; Sun and Abraham, 2021). That repair concerns aggregation rather than identification: these estimators maintain parallel trends and are biased when it fails, which is the failure we address.

estimand decomposes as

$$\beta = \tau + \delta, \quad \tau_{\text{pre}} = 0, \quad (2)$$

where τ collects the causal effects (zero before treatment, by no anticipation) and δ is the differential trend between treated and comparison groups that would have occurred absent treatment. Parallel trends is the restriction $\delta_{\text{post}} = 0$, under which $\beta_{\text{post}} = \tau_{\text{post}}$, and a pre-trends test is a test of $\delta_{\text{pre}} = 0$.

We will require the analogous objects at the cohort level. Write δ_g for the differential trend of cohort g and say that parallel trends holds for g if $\delta_{g,\text{post}} = 0$. A key feature of applications is that this may hold for some cohorts and fail for others. Defending the timing of treatment as unrelated to the outcome path is already delicate in a two-group design, and staggered adoption raises the burden rather than lowering it, since each cohort's timing must be defended on its own: one cohort's adoption may be as good as randomly timed while another's coincides with a confounding shock.

Finally, we distinguish two regimes for how cohorts are judged credible, because they carry different inferential guarantees. Under ex-ante (or independent) selection, credibility is decided from pre-determined covariates, institutional knowledge, or an independent data split, so that the selected set is statistically independent of the estimation errors in $\hat{\beta}$. Under data-driven selection, credibility is assessed from the realized pre-trends $\hat{\beta}_{\text{pre}}$, so that the selected set depends on the estimation sample. The distinction is immaterial for the estimand and its identification, but central for inference, and we treat it explicitly below.

2.2 Estimand and identification

Let S denote a set of cohorts judged to have credible parallel trends, produced by a selection rule R that we specify below. We define the target as the treatment effect for that subpopulation.

Definition 1 (Credible-subpopulation LATT). *For cohort weights $w_g > 0$ (e.g. proportional to cohort size), the credible-subpopulation local ATT is*

$$\theta_S = \frac{\sum_{g \in S} w_g \text{ATT}(g, \cdot)}{\sum_{g \in S} w_g}, \quad (3)$$

where $\text{ATT}(g, \cdot)$ is any within-cohort aggregation of interest (a fixed event-time, or an average over post-treatment periods).

The value of the target is that it is identified under a weaker condition than the ATT.

Proposition 1 (Identification). *If parallel trends holds for every $g \in S$, i.e. $\delta_{g,\text{post}} = 0$ for all $g \in S$, then θ_S is point-identified by the reweighted group-time effects, $\theta_S = (\sum_{g \in S} w_g \beta_{g,\text{post}}) / \sum_{g \in S} w_g$, and this holds irrespective of whether $\delta_{g,\text{post}} \neq 0$ for cohorts $g \notin S$.*

Proof. For $g \in S$, $\beta_{g,\text{post}} = \tau_{g,\text{post}} + \delta_{g,\text{post}} = \tau_{g,\text{post}}$ by the hypothesis and (2). The reweighted sum over S therefore equals the corresponding weighted average of causal effects, which is θ_S by definition. Cohorts outside S receive zero weight, so their differential trends do not enter. $\square$

Proposition 1 is elementary, but it is the paper’s central point. By changing the target to the credible subpopulation, one converts a partially-identified problem (the ATT, when some cohorts violate parallel trends) into a point-identified one, at the cost of answering a narrower question. This is exactly the trade made by Imbens and Angrist (1994), Crump et al. (2009), and Li et al. (2018) in other settings, and it carries the same interpretive obligation: one must characterize the subpopulation S that the estimand describes.

The construction uses staggered timing only to supply the groups. What Proposition 1 requires is a partition of the treated population into groups whose parallel trends can each be assessed, together with the event-study structure that makes such assessment possible. Adoption cohorts are the natural groups under staggered timing, and their credibility heterogeneity comes for free from the variation in when and why units adopt. In a non-staggered design, where all treated units share one treatment date, the same construction applies with groups defined by covariates or geography, provided two conditions that staggering would otherwise supply are argued rather than assumed: the groups must differ in the credibility of their parallel trends, and there must be enough pre-periods to estimate each group’s pre-trend rather than chase noise. The groups should accordingly be subpopulations large enough to assess, not individual units, whose pre-trends are too noisy to select on without reintroducing the pre-test bias in its most severe form. Read this way, the method is the parallel-trends analogue of selecting on overlap (Crump et al., 2009; Li et al., 2018): one trims to the subpopulation on which the identifying assumption is credible, using pre-trends in place of the propensity score.

2.3 Selecting the credible cohorts

The selection rule R maps the data to the set S , and its form fixes both which subpopulation θ_S describes and the inferential guarantees available. It takes two forms, matching the two regimes of Section 2.1. Under ex-ante selection, S is fixed before estimation from pre-determined covariates, institutional knowledge of which cohorts were plausibly confounded, or an independent data split. Under data-driven selection, credibility is read from the estimated pre-trends: a cohort enters S when its estimated pre-trend is close to flat,

$$S = \{ g : \max_{e < 0} |\hat{\beta}_{g,\text{pre}}(e)| \leq c \}, \quad (4)$$

for a threshold c .

This is the direct reading of parallel trends. The assumption is that the treated-comparison difference would have stayed constant absent treatment, so a cohort is credible when its pre-treatment difference is close to constant, and the screen keeps exactly those. It is posed in the same currency as the honest inference of the next subsection, which bounds how far the post-treatment difference could have wandered from flat, so selection and inference test one object: the level of the parallel-trends violation.²

²A researcher willing to trust that a straight pre-trend would have continued linearly can relax flatness to approximate linearity, keeping cohorts whose pre-difference drifts steadily but does not bend and screening on the

The threshold trades scope against credibility: a lax c retains more cohorts but admits larger residual violations, a strict c the reverse. Because a cohort’s pre- and post-treatment coefficient errors are correlated in finite samples, rule (4) makes S random and dependent on the estimation sample, which is the source of the pre-test bias below and of the inference subtleties that follow it.

We do not recommend committing to a single c . Because the sensitivity interval of Section 2 is applied to whichever set c selects and must dominate that set’s residual pre-trend, a stricter c yields a cleaner set that a smaller bound M suffices to cover, while a laxer c buys scope at the cost of a larger M . The threshold therefore trades scope for precision rather than credibility for nothing, and the honest object to report is the estimand together with its interval traced as a function of c , the selection analogue of the breakdown curve we report for M . A reader then sees the scope-credibility-precision frontier directly and need not trust a single cutoff. When a single value is nonetheless wanted, we recommend calibrating it to negligibility rather than to significance: set c to the largest pre-trend that would bias θ_S by less than a stated tolerance, in the spirit of the non-inferiority tests of Bilinski and Hatfield (2020), rather than to a critical value of a pre-trends test, which would reinherit the low power that motivates the paper. The application of Section 4 reports such a path.

2.4 Estimation

The estimator $\hat{\theta}_S$ replaces population objects by their Callaway and Sant’Anna (2021) sample analogues and reweights over the selected cohorts.

Proposition 2 (Consistency and efficiency). *Under ex-ante selection and the regularity conditions of Callaway and Sant’Anna (2021), $\hat{\theta}_S$ is consistent for θ_S . If parallel trends holds for all cohorts, $\hat{\theta}_S$ is consistent for the ATT, with an efficiency loss relative to estimators that use all cohorts.*

Under data-driven selection the rule (4) couples S to the estimation errors: a confounded cohort whose pre-trend noise happens to fall below c is admitted, and its differential trend enters $\hat{\theta}_S$. This is a pre-test selection bias in the sense of Roth (2022), with a definite source and a definite fate.

Proposition 3 (Pre-test bias). *Under data-driven selection, $\hat{\theta}_S$ carries a finite-sample bias equal to the weighted average of the admitted confounded cohorts’ post-treatment differential trends, times their probability of admission. As per-cohort information grows this admission probability tends to zero, so the bias vanishes and $\hat{\theta}_S$ is consistent for θ_S , though not finite-sample unbiased.*

2.5 Honest inference on the selected set

Selecting credible cohorts does not by itself deliver parallel trends: a cohort with a flat pre-trend may still have a nonzero post-treatment differential trend, and selection removes gross offenders but not subtle residual violations. We therefore accompany the point estimate with the sensitivity analysis

pre-trend’s curvature, its second difference, instead. This buys scope at the cost of the extrapolation assumption, and it pairs with the smoothness sensitivity class rather than the level bound below. We develop this refinement in Appendix A.

of Rambachan and Roth (2023), applied to the selected-cohort aggregate: one imposes a restriction $\delta_{S,\text{post}} \in \Delta$ on the residual differential trend and reports the implied confidence set for θ_S . Matching the flatness criterion of the screen, we adopt the level bound $\Delta^{\text{Level}}(M) = \{\delta : |\delta_{\text{post}}(e)| \leq M \ \forall e\}$, which asks how far the treated-comparison difference could have wandered from its flat pre-treatment level, absent treatment. The estimator is then the raw reweighted post-treatment average, with no extrapolation, and the associated fixed-length confidence interval (FLCI) is near-optimal (Armstrong and Kolesár, 2018). Because M is measured in the outcome’s own units, the breakdown value M^* reads directly: it is the size of post-treatment parallel-trends violation that would just overturn the stated conclusion.

Validity depends on the selection regime. Under ex-ante selection S is independent of $\hat{\beta}$, so the uniform validity of Rambachan and Roth (2023) applies verbatim. Under data-driven selection S is random, and two issues arise. The first is post-selection inference: when cohorts are well separated from the threshold, selection is asymptotically harmless.

Proposition 4 (Selection consistency and oracle coverage). *Suppose the statistic on which selection is based is consistent for its population counterpart, and that the following separation condition holds: there exists $\epsilon > 0$ such that, for every cohort, the population value of that statistic lies at distance at least ϵ from the threshold c . Then $P(\hat{S} = S^*) \rightarrow 1$, where S^* denotes the population credible set, and the FLCI computed on the estimated set $\hat{S}$ has the same asymptotic coverage as the oracle interval computed on S^* , and in particular attains its nominal level.*

Proof. By consistency of the selection statistic together with the separation condition, the probability that any given cohort is misclassified tends to zero. Since the number of cohorts is finite, a union bound gives $P(\hat{S} \neq S^*) \rightarrow 0$. On the complementary event the estimator, its covariance estimate and the resulting interval coincide exactly with their oracle counterparts computed on the non-random set S^* . The two sequences therefore share a limiting distribution, and the coverage of the interval computed on $\hat{S}$ converges to that of the oracle interval, which is nominal by the validity of Rambachan and Roth (2023) applied to a fixed set. $\square$

Two caveats attach to Proposition 4. First, the separation condition is untestable, and is nearly the parallel trends question in disguise: to assert that every cohort is plainly credible or plainly not is close to asserting one already knows which cohorts are credible, so the result relocates the identifying content rather than supplying it. Second, the guarantee is pointwise, not uniform: Leeb and Pötscher (2005) show that a post-selection estimator’s distribution cannot be estimated uniformly consistently, so naive post-selection intervals lack uniform validity (Leeb and Pötscher, 2008). The obstruction is confined to local-to-threshold configurations, where the selection statistic drifts toward the boundary at the sampling rate. Away from the boundary, selection is asymptotically deterministic and inference behaves conventionally. A researcher unwilling to assume separation can sidestep both caveats by selecting on information independent of the estimation errors, such as pre-determined covariates, a sample split, or randomized (data-carving) selection (Fithian et al., 2014). These buy

validity through independence between selection and estimation rather than correct selection, at some cost in efficiency.

The second issue is that the bound M must accommodate a random selected set. The FLCI at M is valid only if the selected set’s residual violation lies in $\Delta^{\text{Level}}(M)$. Since that violation is itself random, unconditional coverage requires M to dominate it with high probability.

Lemma 1 (Calibration to the random selected set). *Under data-driven selection, choosing M equal to the mean residual violation of the selected aggregate undercovers. Unconditional coverage requires M to exceed the mean and to approach an upper quantile of the residual-violation distribution, with the FLCI’s sampling slack providing a partial buffer.*

The practical implication is that M should be calibrated to the residual violation of a typical selected set, not to its mean.

Finally, we caution against one otherwise-natural choice of Δ . The relative-magnitudes restriction $\Delta^{RM}(\bar{M})$ of Rambachan and Roth (2023), following Manski and Pepper (2018), bounds post-treatment violations by $\bar{M}$ times the observed maximal pre-treatment violation. Because it anchors its bound to the observed pre-trends, it is self-undermining precisely in the regime where our method’s value is in question.

Proposition 5 (Relative magnitudes is self-undermining). *The identified-set half-width under $\Delta^{RM}(\bar{M})$ is proportional to the observed pre-trend magnitude. When pre-trends are uninformative about post-treatment violations, the observed pre-trends are small while the true violation need not be, so $\Delta^{RM}(\bar{M})$ produces intervals whose width tends to zero even as the bias does not, yielding undercoverage. A fixed restriction such as $\Delta^{\text{Level}}(M)$, whose bound is set by the researcher rather than read off the observed pre-trends, does not share this defect.*

3 Simulation study

3.1 Design

We study the method in two environments. In the reduced-form model we draw event-study coefficients directly from their asymptotic distribution, $\hat{\beta} \sim \mathcal{N}(\tau + \delta, \Sigma)$ with Σ known, which isolates the identification and inference mechanics in the setting where the underlying theory is exact. In the panel model we generate individual outcomes for a staggered design and estimate both the coefficients and their covariance from the data, which confirms that the conclusions survive realistic estimation. Unless stated otherwise, results are averages over eight to twenty thousand replications.

The cohort structure is the same throughout. There are twelve cohorts, of which some are clean, with $\delta_g = 0$, and the remainder confounded, with a nonzero differential trend. Every cohort is given the same true treatment effect, normalized to one. This normalization is deliberate: it makes the true ATT and the true LATT both equal to one, so that any departure of an estimator from one

is attributable to a violation of parallel trends rather than to treatment-effect heterogeneity. It therefore isolates the identification problem that the paper is about.

Throughout, each cohort carries a four-period pre-treatment and four-period post-treatment event study with the reference period normalized to zero. A confounded cohort carries a level violation: its treated-comparison difference shifts by $V > 0$ in the post-period and by ϕV in the pre-period, for a scalar $\phi \geq 0$ we call the informativeness parameter and that is the master axis of the study. When $\phi = 0$ the confound is flat before treatment and no pre-trend rule can detect it; as ϕ grows the pre-period shift announces it and it becomes detectable. Because V is the post-treatment violation it is directly comparable to the level bound M of $\Delta^{\text{Level}}(M)$, which makes the validity statement “covers if and only if $M \geq V$ ” checkable, and it makes ϕ the operational version of the informativeness of pre-trends discussed in Section 2.

The selection rule is the flatness screen, posed in the same currency as the honest inference: cohort g is retained if its estimated pre-trend is close to flat, $\max_e |\hat{\beta}_{g,\text{pre}}(e)| \leq c$. The estimator of the ATT averages the post-treatment effect over all cohorts, as a heterogeneity-robust estimator would, and the LATT averages it over the selected cohorts only. Because the sampling errors of a cohort’s pre- and post-treatment coefficients are correlated, the selection is genuinely data-dependent in the sense of Section 2, and the pre-test bias of Proposition 3 is operative.

3.2 The estimand: recovering a clean effect where the ATT is biased

We first illustrate Proposition 1 in the simplest configuration, with a subset of cohorts violating parallel trends in the post-period. Table 1 reports the result. The estimator of the ATT is biased by +0.267: it averages over all cohorts, so the offending cohorts’ differential trends enter it directly. The LATT lands at 1.000, on the truth, because the selection removes those cohorts from the average and, by Proposition 1, their violations then do not enter the estimand at all.

The lower panel of the table sweeps the sampling noise s , and is the more informative comparison. As precision improves the LATT’s bias falls monotonically from +0.015 to zero: this residual is the pre-test bias of Proposition 3, which arises because the selection is made on the same noisy pre-trends used for estimation, and it vanishes as the selection rule becomes reliable. The ATT estimator’s bias, by contrast, is frozen at +0.267 regardless of precision. The contrast is the paper’s basic point in miniature: the LATT faces a noise problem, which more data cures, whereas the ATT estimator faces a wrong-target problem, which no amount of data cures.

3.3 The scope condition

Figure 1 traces the method across the informativeness axis ϕ described in Section 3.1, and its four panels should be read together.

The upper-left panel plots the bias of the two point estimators against the truth. The bias of the ATT estimator is a flat line at +0.30: it never consults pre-trends, so its performance cannot depend on how informative they are. The LATT’s bias begins on top of that line at $\phi = 0$ and falls to zero as pre-trends become informative. The vertical gap between the two curves is therefore exactly the

Table 1: The estimand demonstration (reduced-form model). True effect = 1 for all cohorts.

	ATT	LATT
Point estimate	1.267	1.000
Bias vs. truth	+0.267	+0.000
Bias as sampling noise s falls:		
$s = 0.40$	+0.266	+0.015
$s = 0.20$	+0.266	+0.001
$s = 0.12$	+0.267	−0.000
$s = 0.03$	+0.267	−0.000

method’s advantage, and reading it off from left to right gives the scope condition: the advantage is nil when pre-trends carry no information about post-treatment violations, and maximal when they carry a great deal.

That the two curves coincide at the left edge is not a defect but an honest reflection of the fact, argued in Section 2, that selecting on uninformative pre-trends buys nothing.

The upper-right panel explains why the LATT’s bias falls, by plotting the identification gap of the selected set, the average post-treatment violation among the cohorts that survive selection. It tracks the LATT’s bias almost exactly, falling from 0.30 to essentially zero. This confirms that the LATT’s remaining bias is not an artifact of the estimator but simply the residual violation that selection has failed to remove. The lower-right panel gives the mechanism behind both: the average number of confounded cohorts that slip through the screen falls from 5.09 of six when the confound is invisible to essentially zero when it is plainly visible.

The lower-left panel turns to inference and reports the coverage of a point-based confidence interval for the causal target, computed both in the naive way and by sample-splitting. Both collapse at low informativeness and recover only at high informativeness. The comparison is diagnostic: sample-splitting removes any contamination from data-dependent selection by construction, so the fact that it collapses just as badly establishes that the failure is not selection distortion but the identification gap of the upper-right panel, since the surviving cohorts genuinely still violate parallel trends. This is what motivates the sensitivity bounds of the next subsection, since no purely inferential fix can repair a violation of the identifying assumption.

(At the right edge the naive interval slightly outperforms the split one. This is simply because splitting discards half the data, and by that point selection is nearly deterministic, so there is no contamination left to correct.)

3.4 Honest inference restores coverage

We now validate the level-bound FLCI of Section 2 applied to the selected-cohort aggregate, using the level violation of Section 3.1 whose size V is directly comparable to the assumed bound M .

Table 2 is the validity check. Read down the rows. When the assumed bound equals the true violation ($V = M$) the FLCI covers at 95.0%, that is, at its nominal level. When the bound is

Master-axis sweep: everything is governed by pre-trend informativeness
($\theta=1$ for all cohorts, so true ATT = true LATT = 1; flatness screen)

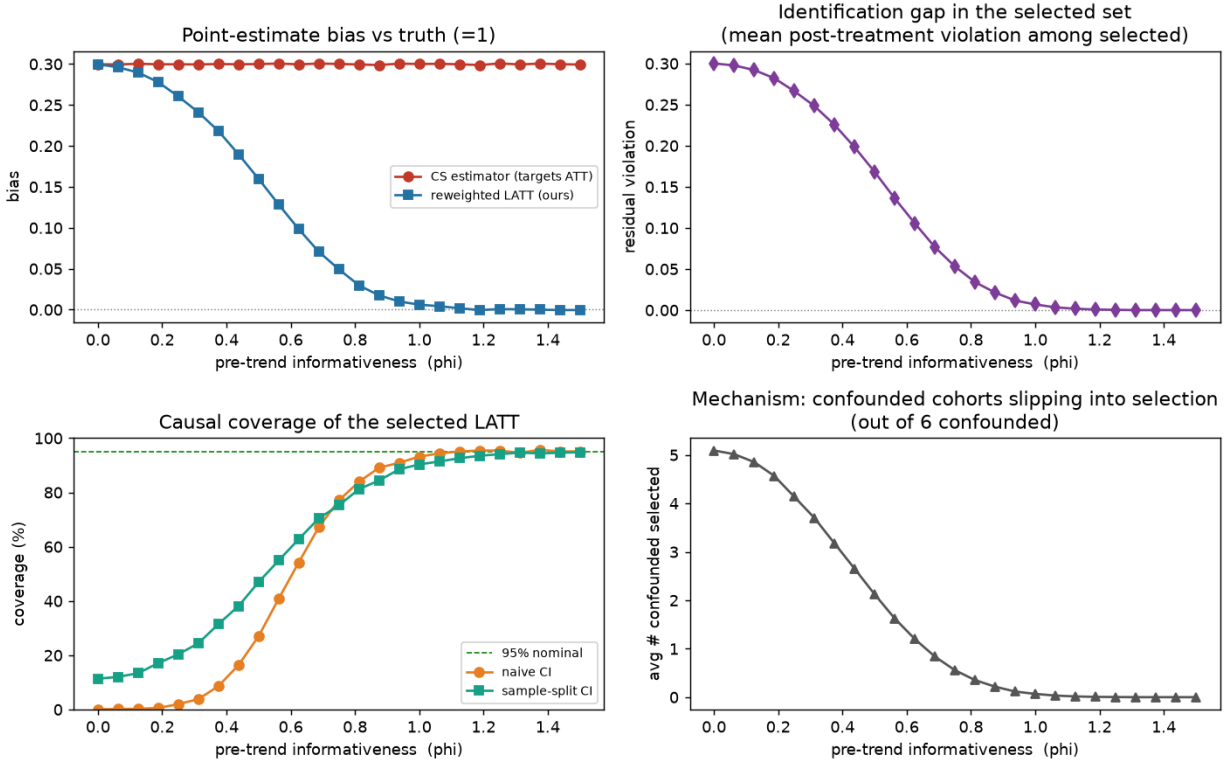


Figure 1: Scope condition. Everything is governed by pre-trend informativeness. The ATT estimator’s bias is constant, and the LATT’s advantage is the gap between the lines. Causal-coverage collapse at low informativeness is an identification gap (surviving cohorts still violate parallel trends), not selection distortion, since sample-splitting collapses identically.

generous ($M > V$) it covers conservatively at 100%. When the bound is violated ($M < V$) coverage collapses. This is precisely the behavior an honest sensitivity interval should display. It is valid exactly on the range of violations it claims to accommodate, and it fails visibly outside it, rather than failing silently.

The final column records that the naive point interval collapses to near-zero coverage the instant any violation is present, because it is centered on a biased estimate with a width reflecting sampling noise alone.

Figure 2 repeats the exercise across a continuum of violation sizes and adds the practical output. In the left panel the point interval falls away rapidly as the violation grows, while each FLCI holds its nominal level until the true violation reaches its own assumed bound, marked by the vertical dotted lines, and declines thereafter. The middle panel records the price: interval half-width is governed by the assumed M rather than by the realized violation, so honesty about a wider class of violations is paid for in width, immediately and proportionately.

Table 2: FLCI coverage of the causal target (reduced-form model): valid iff $M \geq V$.

True violation V	Assumed M	FLCI coverage	Point-estimate coverage
0.30	0.30 ($= V$)	95.0%	15.9%
0.30	0.60 ($> V$)	100%	14.9%
0.60	0.30 ($< V$)	9.3%	0.0%
0.60	0.60 ($= V$)	95.0%	0.0%

Level-bound sensitivity: honest coverage, the price in width, and the breakdown curve

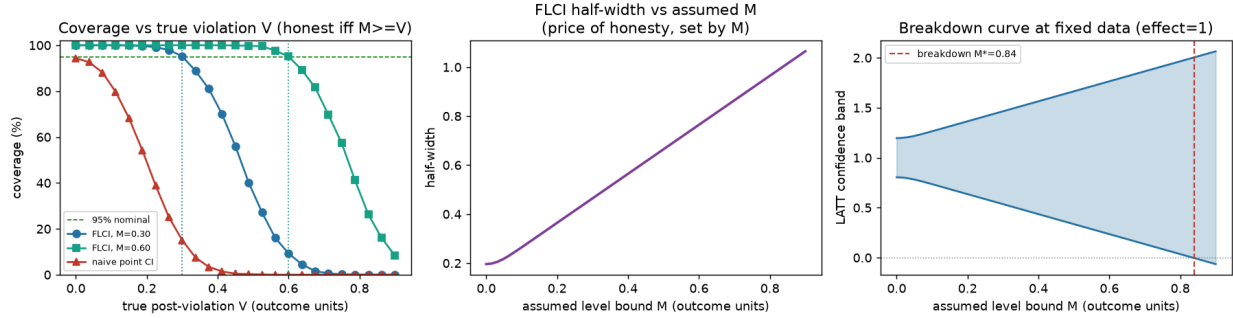


Figure 2: Sensitivity bounds restore honest coverage. The point estimate collapses as the violation grows. The FLCI is honest exactly while $M \geq V$. Interval width is the price of honesty, set by M . The right panel is the sensitivity/breakdown curve, with M^* in outcome units.

The right panel is what a practitioner would actually report. Holding the data fixed, it traces the confidence band as the assumed bound is relaxed and identifies the breakdown value M^* at which the band first admits the truth. Because M is in the outcome’s own units, this quantity answers the question a reader of a DiD paper wants answered directly: how large a post-treatment parallel-trends violation one must be willing to countenance before the stated conclusion no longer follows. And because the bound is fixed by the researcher rather than read off the pre-trends, the exercise remains informative even when pre-trends are uninformative, in direct contrast to the relative-magnitudes restriction of Proposition 5, whose bounds would shrink toward zero in exactly that case and deliver false precision.

3.5 Selection does not distort the bounds, but calibration must respect randomness

Two questions remain for the data-driven regime. Does the fact that the selected set was chosen using the same data distort the sensitivity interval, and how should the bound M be chosen given that the selected set, and hence its residual violation, is random? Figure 3 answers both, and its two panels are best read in reverse order.

The right panel displays the distribution of the residual violation of the selected aggregate across replications. It is visibly spread and multi-modal, its mass running from zero to about 0.06 with a thin tail out to 0.11. Each mode corresponds to a different number of borderline cohorts surviving

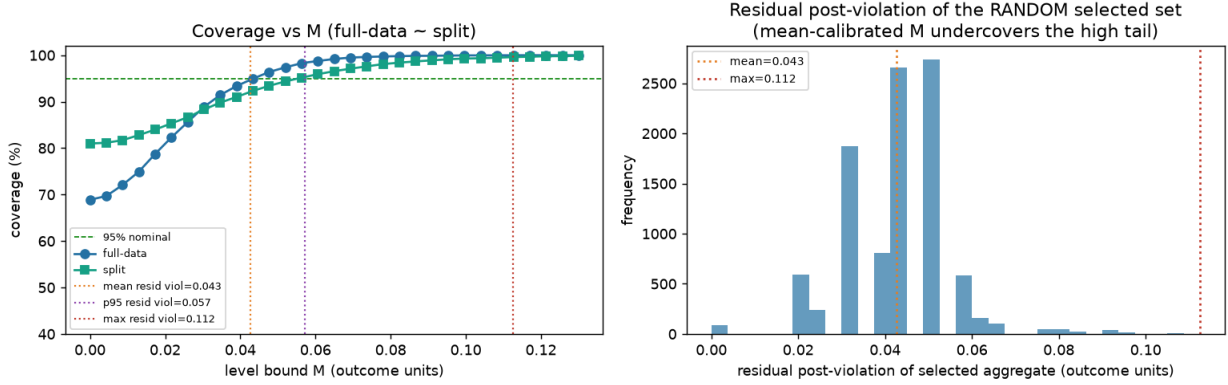


Figure 3: Calibrating the sensitivity bound to the random selected set. Full-data and split coverage cross nominal at nearly the same M (no gross selection distortion). A mean-calibrated M undercovers, with nominal coverage reached near the ninety-fifth percentile of the residual violation.

the screen in that replication. This is the concrete content of Lemma 1: the object that a fixed bound M must dominate is not a constant but a random variable, and a bound chosen to sit at its mean will be exceeded in a substantial share of samples.

The left panel confirms the consequence and settles the distortion question. Coverage is plotted against M for the full-data procedure and for sample-splitting. The two curves lie close to one another and cross the nominal level at nearly the same bound, the full-data procedure at $M \approx 0.048$ and the split at $M \approx 0.056$. Data-dependent selection therefore introduces no gross distortion of the FLCI here, and splitting buys little on this account. The small residual gap reflects splitting's use of half the data, and it is consistent with the pointwise, as opposed to worst-case, reading of Leeb and Pötscher (2005) discussed in Section 2.

Turning to calibration, the vertical reference lines mark the mean, the ninety-fifth percentile and the maximum of the residual-violation distribution. Setting M at the mean, 0.043, leaves coverage below nominal; reaching 95% requires $M \approx 0.048$, near the ninety-fifth percentile 0.057 and well below the maximum 0.112, the shortfall being absorbed by the sampling slack already built into the interval. The practical recommendation follows directly: calibrate the sensitivity bound to the residual violation of a typical selected set, not to its average, and treat the resulting bound as a floor rather than a target.

3.6 Panel confirmation

The exercises above all draw coefficients from their asymptotic distribution with a known covariance, which is the right setting in which to check identification and inference mechanics but is not the setting a practitioner faces. We therefore repeat the estimand demonstration on generated panel data. Individual outcomes are simulated for a staggered adoption design with a level confound and idiosyncratic noise. Cohort-level event-study coefficients are then estimated from genuine

difference-in-differences contrasts against a common never-treated comparison group, screened on flatness exactly as in the reduced-form model, and their covariance is estimated from the data rather than supplied.

The conclusions are unchanged. Across the informativeness axis the ATT estimator’s bias remains flat near +0.30 while the LATT’s bias falls toward zero, reproducing the upper-left panel of Figure 1 with estimated rather than known quantities. The estimated covariance is well calibrated in the region where selection is effectively deterministic, with the estimated standard error matching its Monte Carlo counterpart and coverage near 95% at high informativeness, which confirms that the inference machinery is not being flattered by the reduced-form assumption of a known Σ .

One discrepancy is worth reporting because it is informative rather than incidental. At intermediate informativeness, where selection is genuinely stochastic, the estimated standard error of the LATT understates its Monte Carlo counterpart (about 0.016 against 0.024 at $\phi = 0.75$). The gap is the component of variance induced by the randomness of the selection itself, which a standard error computed conditional on the realized selected set does not capture. This is an independent confirmation, from the panel side, of the inferential issues discussed in Section 2: conditioning on a data-dependent selection understates uncertainty, which is precisely why the honest object to report is the sensitivity interval of Section 3.4 rather than a conventional confidence interval around the point estimate.

4 Empirical application

We illustrate the method with the local effect of the shale boom on house prices, a staggered design with many counties per cohort. The treatment date is the year a county’s oil and gas production first rises sharply, from county production records (U.S. Department of Agriculture, Economic Research Service, 2015), which sorts the boom counties into onset cohorts; the outcome is the log county house-price index (Bogin et al., 2019), over 1998–2019. We estimate Callaway and Sant’Anna (2021) group-time effects against never-treated counties, those with negligible production, screen cohorts on the flatness of their estimated pre-trend, and report the level bound of Section 2 for the selected aggregate.

Onset is gradual and partly endogenous, which is what makes the setting instructive. A county’s production ramps up over several years, and where it does so during a regional housing upswing its prices are already climbing before the boom is dated. Figure 4(a) shows the result: the early cohorts have flat pre-trends and are retained, while the later cohorts, whose onset coincides with the mid-2000s house-price run-up, carry pre-treatment trends several times larger and are dropped.

The two aggregates part ways (Figure 4b). Pooled across all cohorts, the event study climbs through the onset date to a house-price effect of +6.7 log points ($t = 8$), the kind of estimate a practitioner would report as a clear positive effect of the boom. But the pooled path is already trending before treatment, and the effect is inherited from the dropped cohorts: restricted to the credible subpopulation, the event study is flat both before and after onset, and the LATT is -0.2

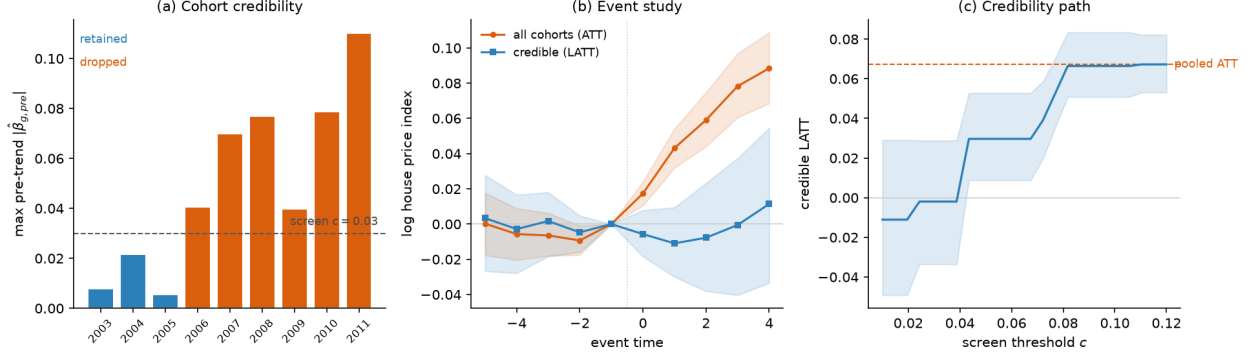


Figure 4: The shale boom and county house prices. (a) Cohort credibility map: maximum pre-treatment violation by onset cohort, with the flatness screen; the later, trending cohorts are dropped. (b) Event study, all cohorts (pooled ATT) versus the credible subpopulation (LATT), with 95% bands. (c) The credibility path: the LATT as the screen threshold c is relaxed, rising from zero toward the pooled ATT.

log points, indistinguishable from zero. The gap between the two, +6.9 log points, is itself sharply estimated ($t = 5$). Panel (c) traces the estimate as the screen is relaxed, the credibility path of Section 2.3: it rises from zero toward the pooled value as the trending cohorts are readmitted, so the positive pooled figure is precisely the contribution of the cohorts whose pre-trends make them least credible. The level bound confirms that this is not a knife-edge of the threshold; the credible estimate stays bounded near zero across it, and the pooled figure cannot be restored without readmitting the non-credible cohorts rather than by positing a residual violation.

The reading is that the boom’s apparent effect on house prices is a pre-existing trend in the counties that adopt late, not a causal effect, and a near-zero net effect is what the offsetting amenity and disamenity channels of local energy development would predict (Muehlenbachs et al., 2015). The application shows the method working in the direction opposite to point-identification: not recovering an effect the pooled estimate misses, but withdrawing one it spuriously reports, and locating that withdrawal exactly in the cohorts the screen flags.

5 Conclusion

We have argued that when parallel trends fails for some treated cohorts, a productive response is to change the estimand rather than to defend or bound the ATT. The credible-subpopulation LATT is point-identified precisely when the ATT is not, is estimated by a transparent reweighting of standard group-time effects, and can be accompanied by honest sensitivity bounds on the residual violation of the selected set. This is a specific instance of the general principle, familiar from instrumental variables and limited overlap, of retreating to a credibly-identified subpopulation.

The method delivers a sharp answer where the ATT estimator is biased and its sensitivity set is wide, at the price of a narrower, subpopulation target whose composition must be described rather than assumed. Its advantage is contingent on pre-trends being informative about post-

treatment violations. Where they are not, only economic context can carry the assumption. Under data-driven selection the estimator carries a vanishing pre-test bias and a pointwise, rather than uniform, guarantee, which a researcher can trade for uniform validity by selecting on independent information. The principal open question is a decision-theoretic characterization of the region in which the point-identified LATT dominates the set-valued ATT, which would turn the scope condition documented here into a formal recommendation.

A The curvature refinement

The main text defines a credible cohort as one with a flat pre-trend and bounds the residual violation by its level. A researcher willing to make a stronger assumption can keep more cohorts. Suppose one is prepared to trust that a differential trend which is a straight line before treatment would have continued as a straight line afterward. Then a cohort whose pre-trend drifts steadily but does not bend is no longer disqualified: the drift can be extrapolated and removed, and only the departure from linearity, the curvature, threatens the estimate. This pairs the smoothness class $\Delta^{SD}(M) = \{\delta : |\delta(e+1) - 2\delta(e) + \delta(e-1)| \leq M\}$ of Rambachan and Roth (2023) with a curvature screen, $\max_e |\hat{\beta}_{g,\text{pre}}(e+1) - 2\hat{\beta}_{g,\text{pre}}(e) + \hat{\beta}_{g,\text{pre}}(e-1)| \leq c$, and replaces the raw post-average with the minimum-variance affine estimator that differences out a linear pre-trend before extrapolating it (Armstrong and Kolesár, 2018). The trade is explicit: one buys scope, the steep-but-straight cohorts the flatness screen discards, at the cost of the linear-extrapolation assumption and a modest variance premium on the estimator.

The same principle that fixes the level screen to the level bound fixes this curvature screen to the smoothness class: the screen should be posed in the same functional the sensitivity class bounds. Level and curvature are not interchangeable, and Figure 5 makes the difference concrete. A “linear” cohort has a steep but straight pre-trend, large in level and zero in curvature, while a “curved” cohort has a smaller level but genuine curvature. Under $\Delta^{SD}(M)$ the linear cohort is benign, since the interval extrapolates its line and removes it, while the curved cohort is the threat. A level screen ranks the linear cohort as the worse offender and the curvature screen ranks the curved one, so the two screens select opposite sets. The right panel plots honest coverage under each: the curvature screen drops the curved cohort and is honest from $M \approx 0$, while the level screen keeps it, needs $M \approx 0.035$, and drops the benign linear cohort for nothing. Screening in the wrong currency pays twice, in precision and in scope. The lesson is symmetric: under the smoothness class one screens on curvature, exactly as, under the level bound of the main text, one screens on flatness.

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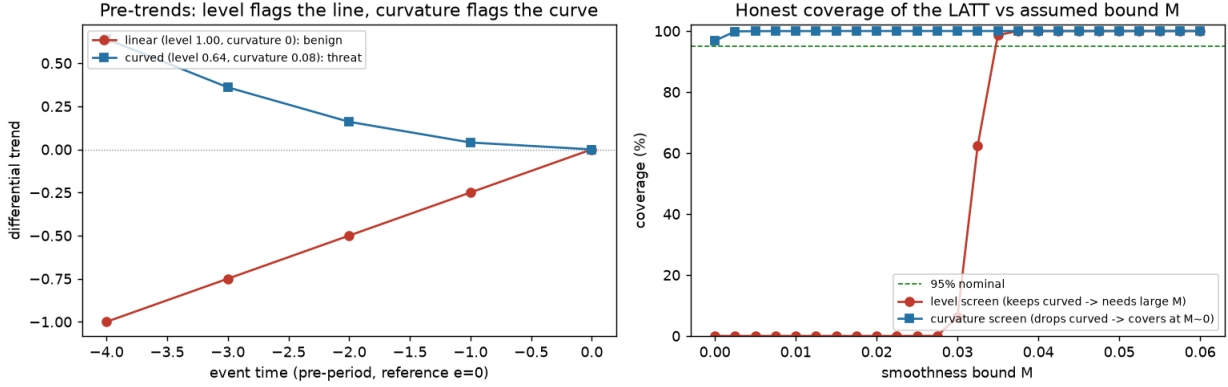


Figure 5: Level and curvature select opposite sets. Left: a steep-but-straight pre-trend (large level, zero curvature) is benign under $\Delta^{SD}(M)$, while a gentler curved pre-trend is the threat. Right: honest LATT coverage versus the assumed bound. The curvature screen drops the curved cohort and is honest at $M \approx 0$, while the level screen keeps it, needs $M \approx 0.035$, and drops the benign linear cohort.

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